10/8/26

Experiment - 5

Packet Capture through Wireshark

AIM:

Experiments on packet capture through wireshark

Packet Sniffers:

→ Sniffs messages being send / received from / by your computer.

→ Store and display contents of various protocol fields in the message.

→ Passive program
- never sends packet itself
- No packet address to PT
- receives a copy of all packets (sent / received)

Packet Sniffer Diagnostic tools

→ Tcpdump
- Eg: tcp dump -exa host 10.129.41.2 -w exe3.out

→ Wireshark
- wireshark -v -exe J.out

Description

Wireshark

A network analysis tool formerly known as Ethereal captures packets in real time and display them in human readable format. Wireshark includes filters, color coding and other features that let you dig deeper.

features

→ Capture network traffic
→ decode packet protocols using dissectors
→ Define filters - capture and display
→ Watch smart stats

5. Create a filter to display only ICMP/ICMPv6 packets and inspect the packet procedure

→ Select total area Connection in Wireshark
→ Go to capture option
→ Select stop capture automatically after 10 0 packets.
→ Then click start capture.
→ Search ICMP/IP packets in search box

6. Create a filter to display only DHCP packets procedure

→ Select LAN Area Connection or Wireshark
→ Go to Capture option
→ Select stop capture
→ Then click start capture
→ Search DHCP packets in search box
→ Save the packets.

2. Create a filter to display only ARP packets and inspect the packets procedure

→ Go to capture option
→ select stop capture automatically after 100 packets.
→ Search ARP packets in search bar.
→ Save the packets.

3. Create a filter to display the DNS packets and provide the flow graph procedure.

→ Go to capture option
→ Select stop capture automatically after 100 packets.
→ then click start capture
→ Search DNS packets in search bar.
→ To see flow graph click statistics flow graph.
→ Save the packets

4. Create a filters to display only HTTP packets and inspect the packets procedure.

→ Select local area connection in wireshark.
→ Go to capture option.
→ Select stop capture automatically after 100 packets
→ Search HTTP packets on search bar.
→ Save the packets

RESULT:

Thus the experiment on packet capture through wireshark is success.

If you're trying to inspect something specific, such as the traffic a program send when phoning home. It helps to close down all the other application using the network. So you can narrow down the traffic. Still you'll likely have a large amount of packets to sift through.

Wireshark is an extremely powerfull tool and ws tutorial is just scratching the surface of what you can do with it professionals use it to debug network protocol implementation execution security problems and inspect network protocol internals.

Capturing and Analyzing packets using wireshark tool.
to filter, capture, view packets in wireshark tool.
Capture two packets from the Ethernet. IEEE 802.3.2 LAN interface and save it.

Procedure

→ Select Local Area connection in network.
→ Go to capture option.
→ Select stop capture automatically after 100 packets
→ Then click start capture
→ Save the packets.

The packet list pane

It displays all the packets in current capture file. The packet list pane each line in the packet list corresponds to one packet in capture file. If you select a line in this pane more details.

The packet Details

The packet details pane shows the current packet in a more detailed form this pane shows protocols and protocol field of the packet selected in packet list.

The Packet bytes

The packet bytes pane shows the data of current packet in hexdump style.

Color coding

You'll probably see packets highlighted in a variety of different colors. wireshark uses colors to help you identify types of traffic and black identifies packet with errors.

Result:

So the experiment on packet capture through wireshark is successfully.

31/8/20 10/10